



RS001 Week 4 — English Worksheet

Topic: Parallel Lists — A Shop Scene · **Course:** Text Adventure (Python) · **Time:** about 45 minutes

This week you use **parallel lists** — several lists that line up by index. Item `[1]`'s name, power, and price all sit at index `1` in three different lists. Together they build a shop.

```
shop_weapons = ["wooden stick", "iron sword", "magic staff"]
shop_power   = [2, 8, 15]
shop_price   = [5, 30, 80]
game.say(f"{shop_weapons[1]} - power {shop_power[1]}, {shop_price[1]} gold")
```

1 · Predict 🧠

Read each piece of code. Before you run it in your head, write what you think the screen will show.

```
names = ["dagger", "axe", "spear"]
price = [10, 40, 25]
game.say(f"{names[1]} costs {price[1]} gold")
```

Which item and which price get printed together?


```
names = ["dagger", "axe", "spear"]
power = [3, 9, 6]
game.say(f"{names[0]} - power {power[2]}")
```

The indexes do not match. What strange sentence comes out?


```
names = ["dagger", "axe"]  
price = [10, 40]  
game.say(f"{names[2]} costs {price[2]} gold")
```

Each list only has 2 items. What happens when the code asks for **[2]** ?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, fix it, then explain why the original was wrong.

Bug A — This shop line should show the **axe** with **its own** price. Right now it shows the axe's name but the dagger's price.

```
names = ["dagger", "axe", "spear"]
price = [10, 40, 25]
game.say(f"{names[1]} costs {price[0]} gold")
```

Hint: the name and the price should share the same index.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A new weapon **"hammer"** was added to the names list, but the other lists were not updated. Now line 4 crashes when it reaches the hammer.

```
names = ["stick", "sword", "staff", "hammer"]
power = [2, 8, 15]
price = [5, 30, 80]
game.say(f"{names[3]} - power {power[3]}, {price[3]} gold")
```

Hint: if one list grows, the parallel lists must grow too.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should print a full 3-line shop menu. Right now line 2 forgot to change its index, so the same weapon shows up twice.

```
names = ["stick", "sword", "staff"]
price = [5, 30, 80]
game.say(f"1. {names[0]} - {price[0]} gold")
game.say(f"2. {names[0]} - {price[1]} gold")
game.say(f"3. {names[2]} - {price[2]} gold")
```

Hint: look closely at the name index on line 2.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · 🎯 Build Your Shop

Open your game. After your opening and inventory, add a **shop scene** built from parallel lists: weapon names, their power, and their price. Print a clear numbered menu the player could buy from. Put it in your hometown if you like.

When it works, send a **photo or video**, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I made three lists for ...

They line up by index so that ...

The item at index 2 was ...

When I added a new weapon, I remembered to ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while your shop prints. Teach it like the viewer has never coded. Try to use these words: **parallel list**, **index**, **power**, **price**, **line up**.

1. Run your game and show the shop menu.
2. Pick one weapon and read its name, power, and price.
3. Point to where all three sit at the same index.
4. Explain what would break if one list were shorter than the others.

Write what you will say in your video. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.